

Portfolio Allocation with a Carbon Objective

EUR Region • Scope 1 + Scope 2

Out-of-sample period: January 2014 – December 2025 • Universe: 470–602 European firms

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Abstract

We construct and evaluate carbon-aware equity portfolios for the EUR region over the out-of-sample period January 2014 to December 2025. Part I builds a rolling long-only minimum-variance (MV) portfolio from a universe of 470 to 602 European firms with complete carbon data, using a 10-year window of monthly returns to estimate a pairwise-complete sample covariance matrix; the matrix is projected onto the positive semi-definite (PSD) cone before optimisation. Part II adds carbon constraints and compares three strategies: a carbon-constrained MV portfolio for the active investor ($MV(0.5)$), a tracking-error-minimising variant for the passive investor ($VW(0.5)$), and a net-zero glide path ($VW(NZ)$) reducing carbon footprint by 10% per year from the 2013 VW baseline.

Key findings: (i) the long-only MV portfolio lowers annualised volatility from 15.87% to 13.65% but loses on risk-adjusted returns relative to the VW benchmark (Sharpe 0.211 vs 0.406); (ii) halving the MV portfolio’s carbon footprint costs essentially zero Sharpe (0.209 vs 0.211) because the heaviest EUR emitters are already excluded by variance minimisation – the constraint is binding in all 12 years yet reshuffles only within the defensive low-volatility segment; (iii) the passive TE-minimisation strategy reduces CF by 50% every year at a realized tracking error of 0.43%, an information ratio of -0.13 and an annual return drag of approximately 6 basis points (Sharpe 0.404 vs 0.406); (iv) the net-zero glide path delivers Sharpe 0.407 (statistically indistinguishable from the benchmark) with realized TE of 0.38% and IR of -0.01 , but the constraint is *not* binding in four years (2014, 2015, 2018, 2021) because the EUR benchmark’s own decarbonisation already met the glide path target; (v) by 2024 the VW(NZ) portfolio reaches 58.7 t CO₂e/M\$ invested, a 71.8% reduction from the 2013 baseline, while the VW benchmark itself declined to 86.6 (a 58% reduction). All results are specific to the EUR region and the chosen sample period.

1 Introduction

The transition to a low-carbon economy has become a central challenge for institutional investors. In the wake of the Paris Agreement and the European Green Deal, asset managers face growing pressure from regulators, clients, and civil society to disclose and manage the carbon footprint of their portfolios. The practical question – how to decarbonise a portfolio without sacrificing financial performance – remains open. Can an investor reduce emissions while preserving risk-adjusted returns? And does the answer depend on whether the investor is active or passive?

This report addresses these questions using 470 to 602 European firms with carbon data over January 2014 to December 2025. The European equity market provides a relevant testing ground: it combines carbon-intensive industries (construction materials, utilities, energy) with a growing share of low-emission sectors (healthcare, technology, financial services). This diversity makes it possible to test whether reducing carbon emissions necessarily comes at a financial cost, or whether both objectives can be achieved simultaneously.

We find that the answer depends critically on the investor’s mandate. For an active investor targeting minimum variance, the decarbonisation constraint has almost no additional cost. The stocks that variance minimisation would normally avoid – cyclical industrials, commodity-sensitive names – are often the same ones that contribute most to carbon exposure. Cutting emissions by 50% requires only marginal portfolio adjustments and leaves the Sharpe ratio essentially unchanged. For a passive investor replicating the value-weighted benchmark the outcome differs: large-cap emitters remain in the portfolio because of their size, so reducing their weight generates a persistent (though small) tracking error.

These results are relevant to the broader discussion on the cost of responsible investing. They also offer a practical insight: the financial penalty of decarbonisation is not fixed – it is a function of the investment strategy, the asset universe, and the baseline against which the emission reduction is measured.

2 Data and Investment Universe

2.1 Data Description

The dataset is sourced from Datastream, covering 1999–2025, and contains 2,545 firms globally with information on prices, market capitalisation, revenues, and carbon emissions. Our analysis focuses on the European region, which comprises 633 ISINs in the Static file. After applying all eligibility filters, the investable universe ranges from 470 to 602 firms depending on the year.

Dataset	Dimensions	Description
Static (ISINs)	(2,545, 4)	ISIN, company name, country, region
RI Monthly	(2,609, 316)	Total return index, monthly, USD
MV Monthly	(2,609, 316)	Market capitalisation, monthly, USD millions
RI Annual	(2,609, 29)	Total return index, annual, USD
MV Annual	(2,609, 29)	Market capitalisation, annual, USD millions
CO ₂ Scope 1	(2,609, 29)	Direct emissions, tonnes CO ₂ e, annual
CO ₂ Scope 2	(2,609, 29)	Indirect emissions, tonnes CO ₂ e, annual
Revenue	(2,609, 29)	Annual revenue, USD thousands
Risk-Free Rate	(312, 2)	Monthly RF factor

Table 1: Raw datasets. Monthly data: December 1999–December 2025. Annual carbon data: 2002–2024. Revenue in USD thousands; market cap in USD millions; CO₂ in tonnes CO₂e.

Monthly observations are used to estimate financial inputs (expected returns and covariance

matrices); annual data drive portfolio construction, rebalancing, and carbon-intensity calculations. Portfolio weights are updated once per year using information available at the end of year Y and are held during year $Y + 1$.

Carbon data coverage is limited before 2010, so the allocation exercise starts in December 2013, using a 10-year estimation window from 2004 to 2013. The investment universe grows over the sample mainly because more firms begin reporting carbon data, rather than because new firms enter the market.

2.2 Data Preprocessing

The raw Datastream output is cleaned before building the portfolio so the investment universe is reliable and suitable for optimisation. We first remove observations that cannot be reliably matched across datasets: rows with missing ISINs are dropped, Datastream error entries are excluded, and firms outside the EUR region are removed. This ensures all datasets are consistent and avoids matching errors downstream.

2.3 Data Cleaning

To address very low prices, total return indices (RI) strictly below the `LOW_FLOOR` of 0.5 are set to NaN. Extremely low price levels can mechanically generate large or unstable returns when computed with very small denominators, so treating them as missing avoids distortions.

To address internal gaps, missing RI values located between two valid observations are forward-filled along the time axis using the last available price. This assumes that temporary reporting gaps do not reflect a fundamental change in firm value and avoids look-ahead bias because future prices are never used. The procedure is bounded: it is only applied between the first and last valid observation for each firm; leading and trailing missing values are left unchanged.

To address delistings, a firm is treated as delisted when it has a valid price at month $t-1$ and a missing price at month t that marks the end of its series. In this case a -100% return is recorded at t and all later months are set to missing. This reflects the economic loss faced by an investor holding the stock at delisting; without it portfolio performance would be biased upward.

To address stale prices and potential liquidity issues, firms are excluded when more than 30% of their monthly returns in the 10-year estimation window are effectively equal to zero (defined as $|R| < 10^{-10}$). A high proportion of zero returns may reflect illiquidity, infrequent trading, or reporting frictions rather than genuine price stability, which would artificially reduce volatility and distort the covariance matrix.

For carbon data, annual CO₂ emissions (Scope 1 and Scope 2) and revenues are forward-filled using the last available observation up to 2024. Missing values at the beginning of the sample – before a firm first reports – are not filled; firms only enter the investment universe once carbon and revenue data are available.

Cleaning step	Count
Prices below 0.50 floor $\rightarrow$ set to NaN	687
Internal gaps forward-filled	147
Firms with potential delisting identified	66
-100% delisting returns applied	66

Table 2: Summary of cleaning operations applied to the raw Datastream dataset.

2.4 Investment Universe Construction

At the end of each year Y , the firms eligible for year $Y+1$ are defined by applying four filters successively to all EUR firms in the Static file. All filters use information available up to December of year Y , avoiding look-ahead bias.

A firm is eligible if it simultaneously satisfies the following criteria:

- It is classified as part of the EUR region in the Static file.
- Its total return index is non-missing and above 0.50 at December year-end.
- It has at least 36 valid monthly return observations over the 10-year estimation period ending December Y .
- Zero-return months represent no more than 30% of the estimation period.
- Both Scope 1 and Scope 2 CO₂ emissions are available for year Y after forward-filling, so the same universe applies to both financial and carbon optimisations.

The parameters in Table 3 are fixed before the simulation and define the universe shown in Table 4.

Parameter	Value	Rationale
REGION	EUR	Group assignment
START_YEAR	2013	First allocation decision (Dec 2013 for 2014)
END_YEAR	2024	Last allocation decision (Dec 2024 for 2025)
ESTIM_YEARS	10	Rolling window = 120 monthly observations
MIN_OBS	36	Minimum 3 years of valid returns
STALE_THR	0.30	Max fraction of zero-return months
LOW_FLOOR	0.50	RI values below this treated as NaN

Table 3: Predefined parameters used to construct the investment universe.

2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
470	488	506	513	526	547	574	593	602	592	577	565

Table 4: Number of EUR firms in the investment universe by year.

3 Estimation and Rebalancing Method

At each annual rebalancing date from 2013 to 2024, expected returns are estimated using the previous 120 months of returns. For each firm, the expected return is the average of its available monthly simple returns:

$$\hat{\mu}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} R_{i,t},$$

where T_i is the number of non-missing observations. This corresponds to the maximum-likelihood estimator under i.i.d. monthly returns. Missing observations are not replaced by zeros: each firm's expected return is computed only from available data, which avoids creating artificial return patterns.

For the covariance matrix, each pair (i, j) is compared using only months in which both firms have valid returns. Firm-level variances are computed separately using each firm's full available history. The covariance between i and j is then:

$$\widehat{\text{Cov}}(i, j) = \rho_{ij}^{\text{common}} \hat{\sigma}_i^{\text{full}} \hat{\sigma}_j^{\text{full}},$$

where $\rho_{ij}^{\text{common}}$ is the correlation estimated on the months common to both firms. This pairwise-complete approach makes better use of the available data than dropping every firm-month with any missing pair.

For rebalancing logic, at the end of each year Y the eligible universe is fixed, the covariance matrix is estimated using the previous ten years of monthly returns, and the optimal weights α_Y are computed. These weights are used during year $Y+1$. Within each year the portfolio is not rebalanced monthly; weights are allowed to drift with realised returns:

$$\alpha_{i,t+1} = \alpha_{i,t} \frac{1 + R_{i,t+1}}{1 + R_{p,t+1}}.$$

Stocks that outperform gain weight; underperformers lose weight. The notebook updates the weight vector after each monthly return and saves the drifted weights as the warm start for the next year's optimisation. The first year uses equal weights as the initial guess. This does not change the optimisation problem but improves stability and realism, since a real investor rebalances from the current portfolio rather than from a flat allocation.

The value-weighted benchmark follows a different rebalancing rule, described in Section 4.2.

4 Part I – Standard Portfolio Allocation

4.1 Minimum-Variance Portfolio

At the end of each year Y we construct a long-only global minimum-variance portfolio. The allocation determined at year-end Y is implemented during year $Y+1$, and performance is measured using realised monthly returns.

The optimisation problem is

$$\min_{\alpha_Y} \alpha_Y' \Sigma_Y \alpha_Y \quad \text{s.t.} \quad \alpha_Y' \mathbf{1} = 1, \quad \alpha_{i,Y} \geq 0 \quad \forall i,$$

where α_Y is the vector of portfolio weights, Σ_Y is the covariance matrix at the end of year Y , and $\mathbf{1}$ is a vector of ones. The full-investment constraint allocates all capital; non-negativity imposes a long-only portfolio, which is important for economic interpretation and especially for the carbon analysis in Part II (ownership weights must be positive).

The covariance matrix is estimated from the previous 120 monthly observations using pairwise-complete handling of missing values. With around 600 firms and only 120 observations, the raw matrix is typically rank-deficient and can be slightly indefinite under pairwise-complete estimation. To ensure the problem is well-posed we project the raw matrix onto the PSD cone by setting negative eigenvalues to zero. This preserves the positive eigenvalues without introducing a shrinkage parameter.

The optimisation is solved with Sequential Least-Squares Quadratic Programming (SLSQP) via `scipy.optimize.minimize`. The objective is the estimated portfolio variance $\sigma_{p,Y}^2 = \alpha_Y' \Sigma_Y \alpha_Y$ and the analytical gradient $\nabla_{\alpha} \alpha_Y' \Sigma_Y \alpha_Y = 2 \Sigma_Y \alpha_Y$ is supplied to the solver. The solver imposes lower bounds of zero, the full-investment equality constraint, a tight convergence tolerance `ftol=10-10`, and a maximum of 1,000 iterations.

4.2 Value-Weighted Benchmark

To evaluate the financial performance of the minimum-variance portfolio we compare it with a value-weighted benchmark $P^{(vw)}$ built on the same investment universe. The benchmark is passive: each firm receives a weight proportional to its market capitalisation. At the end of each

month t ,

$$w_{i,t}^{(vw)} = \frac{\text{Cap}_{i,t}}{\sum_{j=1}^N \text{Cap}_{j,t}},$$

and the monthly return of the benchmark is

$$R_{t+1}^{(vw)} = \sum_{i=1}^N w_{i,t}^{(vw)} R_{i,t+1}.$$

The benchmark serves as a natural reference because it approximates the aggregate market portfolio within the eligible universe.

4.3 Results and Portfolio Comparison

Table 5 reports the annual output of the MV optimisation. The investment universe ranges from 470 firms in 2013 to a peak of 602 in 2021. Despite this breadth, the MV portfolio is highly concentrated, with only 20–32 active positions per year. This concentration is a known feature of constrained long-only MV optimisation, which selects a small subset of low-volatility, low-covariance names.

Year	Firms	Non-zero weights	Ex-ante σ
2013	470	24	9.54%
2014	488	20	8.97%
2015	506	21	9.52%
2016	513	21	9.85%
2017	526	20	9.96%
2018	547	28	9.49%
2019	574	25	8.65%
2020	593	31	8.78%
2021	602	32	8.58%
2022	592	29	9.12%
2023	577	26	9.14%
2024	565	25	9.05%

Table 5: Annual minimum-variance optimisation summary. Ex-ante σ is the optimised portfolio’s expected annualised volatility (using Σ after PSD projection).

The MV portfolio is compared with the VW benchmark $P^{(vw)}$. Benchmark weights are computed from end-of-prior-month market capitalisations so the return in month $t+1$ uses only information available at month t , avoiding look-ahead bias.

Portfolio	Arith. (%)	Geom. (%)	Vol (%)	Sharpe	Min (%)	Max (%)	MDD (%)
$P^{(vw)}$	8.19	7.16	15.87	0.406	−15.34	18.27	−28.17
$P^{(mv)}$	4.62	3.76	13.65	0.211	−12.02	14.03	−23.92

Table 6: Out-of-sample performance statistics, January 2014 to December 2025. Sharpe is computed using the monthly risk-free rate.

As Table 6 shows, the MV portfolio achieves its primary objective: volatility falls from 15.87% (VW) to 13.65%. Downside protection also improves – maximum drawdown is −23.92% vs −28.17% and the worst monthly return is −12.02% vs −15.34%.

However, this comes at a real performance cost. Annualised arithmetic return drops from 8.19% to 4.62%, and Sharpe falls from 0.406 to 0.211. The strategy reduces volatility but does not improve risk-adjusted performance.

Three factors explain this underperformance. First, the portfolio is highly concentrated (20–32 names out of > 500 eligible). Second, MV naturally favours defensive, low-volatility stocks that lag during strong equity rallies. Third, the covariance matrix is estimated from only 120 monthly observations while the universe has roughly 600 firms; even after PSD projection, estimation error affects optimal weights.

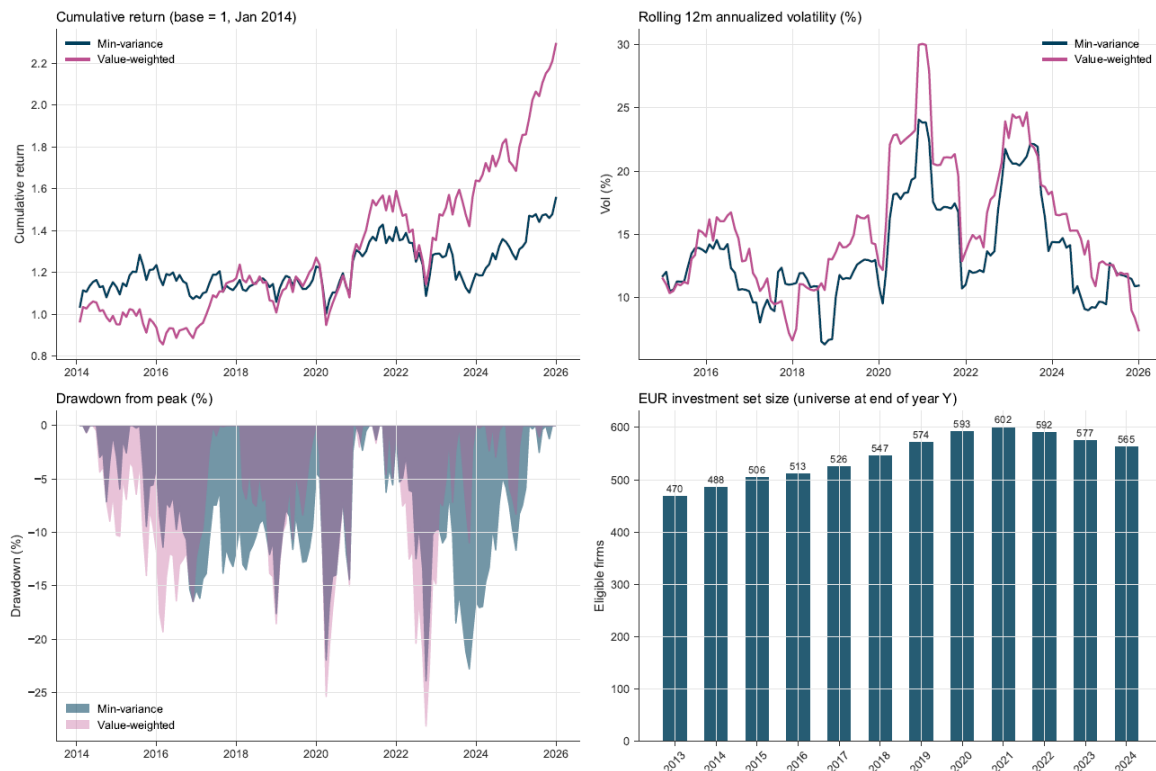


Figure 1: Minimum-variance versus value-weighted portfolio, 2014–2025. Left panel: cumulative return (base = 1 at Jan 2014). Right panel: 12-month rolling realised volatility, annualised.

Figure 1 confirms that MV achieves lower and smoother realised volatility over 2014–2025, but the value-weighted portfolio clearly delivers higher cumulative return.

5 Part II – Portfolio Allocation with Carbon Emission Reduction

5.1 Objective

This part extends Part I by incorporating carbon constraints. We first measure the carbon exposure of the two baseline portfolios (MV and VW) without any constraint. Then we build two decarbonised variants: $MV(0.5)$ for the active investor (minimum variance with a 50% CF reduction relative to the unconstrained MV) and $VW(0.5)$ for the passive investor (tracking-error minimisation with a 50% CF reduction relative to VW). Finally we implement a net-zero glide path $VW(NZ)$ where the allowed CF declines by 10% per year from the 2013 VW baseline. Throughout, we compare financial performance and carbon outcomes across strategies.

5.2 Carbon Metrics: Carbon Intensity, WACI, and Carbon Footprint

We use two complementary portfolio-level carbon metrics. The Weighted-Average Carbon Intensity (WACI) captures the operational carbon intensity of the companies held in the portfolio, relative to their revenues. The Carbon Footprint (CF) measures the emissions attributed to the

investor through ownership stakes. Before computing these we define firm-level Carbon Intensity (CI), the building block of WACI.

At the firm level, CI measures tonnes of CO₂e per million USD of revenue (Scope 1 + Scope 2):

$$CI_{i,Y} = \frac{E_{i,Y}^{S1} + E_{i,Y}^{S2}}{\text{Rev}_{i,Y}/1000}.$$

Since revenues in Datastream are in thousands of USD, we divide by 1,000 to express emissions per million USD of revenue. While a large firm may have high absolute emissions, CI captures emission intensity relative to revenue, allowing comparison across firms of different sizes.

WACI aggregates CI to the portfolio level through portfolio weights:

$$\text{WACI}_Y^{(p)} = \sum_{i=1}^N \alpha_{i,Y} CI_{i,Y}.$$

It captures how carbon-intensive, on average, the companies in the portfolio are relative to their revenues.

CF shifts the perspective from firms' operating efficiency to the investor's ownership responsibility:

$$\text{CF}_Y^{(p)} = \sum_{i=1}^N \alpha_{i,Y} \frac{E_{i,Y}}{\text{Cap}_{i,Y}},$$

where $E_{i,Y}$ is total Scope 1 + Scope 2 emissions, $\text{Cap}_{i,Y}$ is market capitalisation, and $\alpha_{i,Y}$ is the portfolio weight.

The ratio $E_{i,Y}/\text{Cap}_{i,Y}$ captures emissions per dollar of market capitalisation, and the weight determines how much of each firm's emissions is attributed to the investor. We use CF as the binding constraint because the goal is to reduce emissions financially associated with the portfolio. Unlike WACI – based on revenue intensity – CF depends directly on ownership exposure: a firm may have very high CI but a limited impact on portfolio CF if its market cap or weight is small; conversely a large-cap firm with moderate emissions intensity may contribute substantially.

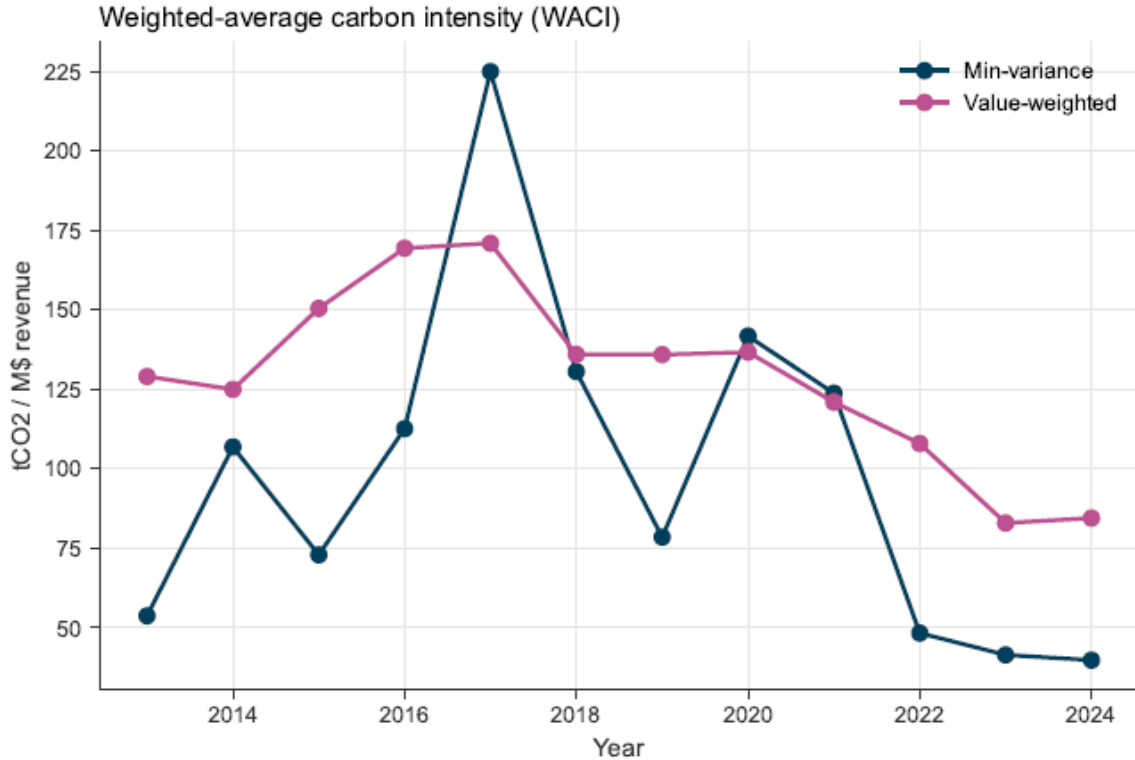


Figure 2: WACI of the MV and VW portfolios, 2013–2024 (tCO₂e per M\$ revenue).

Figure 2 illustrates the carbon profiles of the two baselines. The VW benchmark shows a relatively smooth pattern: WACI peaks at 171.00 tCO₂e/M\$ revenue in 2017 and declines steadily to 84.55 in 2024. This gradual decrease reflects the evolution of the European equity universe – lower reported Scope 1 and Scope 2 emissions for many firms, and the rising weight of less carbon-intensive sectors such as technology and healthcare.

The MV portfolio shows a much more unstable carbon profile. WACI rises from 53.78 in 2013 to 225.00 in 2017, then falls sharply to 39.84 in 2024. This pattern is driven by portfolio construction rather than by an explicit carbon objective: because MV selects names based only on their contribution to portfolio risk, the portfolio can become exposed to carbon-intensive sectors when those firms also display low volatility or low correlation with the rest of the portfolio. Regulated utilities or industrial firms can be attractive from a risk-minimisation perspective but significantly raise WACI; when they leave the optimal allocation in later years, WACI falls again.

By 2022–2024 the relationship reverses: MV WACI is clearly lower than VW (39.84 vs 84.55 in 2024). This does not result from any explicit carbon objective. Rather, in the later years the optimiser favours telecom operators, insurers, and medtech companies, which structurally have lower emissions. This overlap between low-risk and low-carbon characteristics is important for what follows: it helps explain why the carbon constraint in the next section has only a limited impact on the MV portfolio's financial profile.

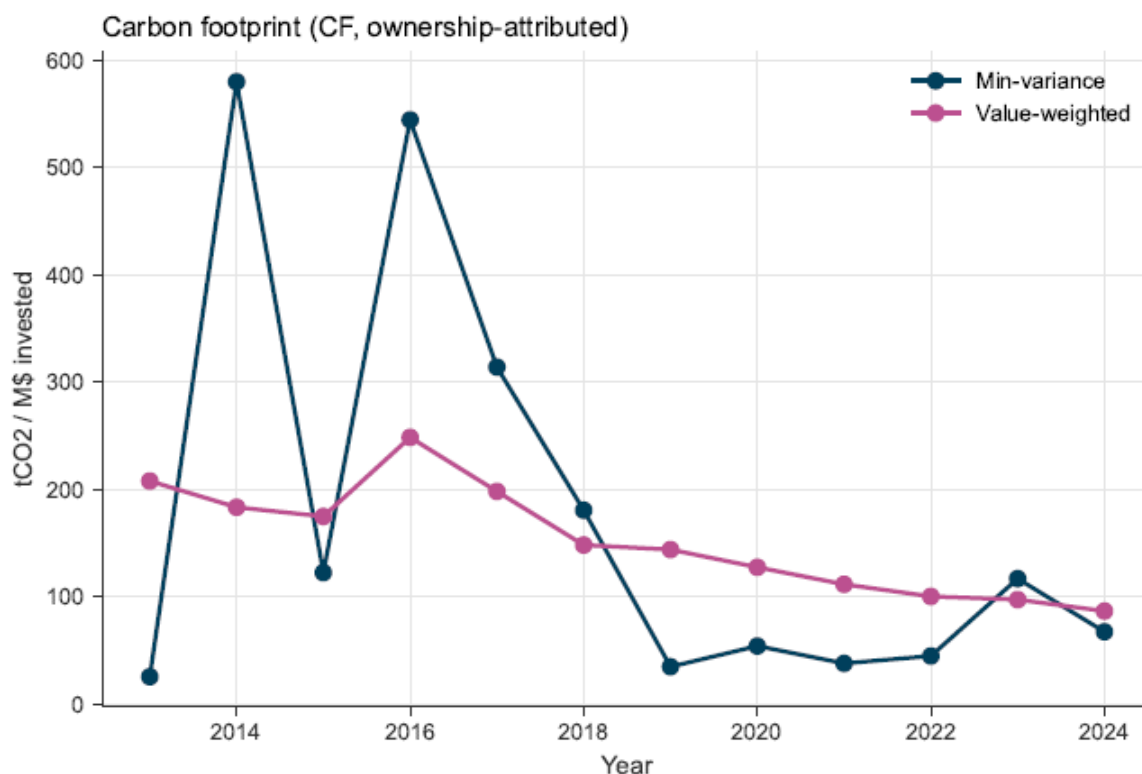


Figure 3: Carbon Footprint of the MV and VW portfolios, 2013–2024 (tCO₂e per M\$ invested).

The CF comparison (Figure 3) is even sharper. The VW benchmark declines smoothly from 207.92 in 2013 to 86.64 in 2024, a 58% reduction. This stable downward trend reflects the diversification of the market-cap-weighted portfolio: weights spread across many firms, no single company has disproportionate influence on aggregate CF.

The MV portfolio reaches CF of 580.15 in 2014 – more than three times the VW level that year – then falls sharply to 122.49 in 2015 and rises again to 544.62 in 2016. These large movements are explained by the concentrated nature of the allocation. In some years the optimiser assigns meaningful weights to utilities or infrastructure firms whose equity returns are stable but whose

emissions are large relative to their market cap. After 2018 the MV CF drops below the VW benchmark and stays there. By 2024 it stands at 67.36 vs 86.64 for VW – a cleaner outcome, but one that arrived by accident. Together, Figures 2 and 3 make the point concrete: without an explicit constraint, an MV investor has no more control over their carbon footprint than a passive index investor, and in some years considerably less.

Top emitters: by intensity vs by ownership impact. Table 7 shows the top-10 most carbon-intensive firms in the EUR universe at three snapshot dates (2013, 2018, 2024). The set of top emitters by CI changed substantially over the sample. In 2013 the list included Drax Group (CI 7,356), Fortum, Endesa, and several construction-materials and energy names – many with non-trivial benchmark weights. By 2024 the average CI of the top-10 had fallen, several historical emitters had dropped out (Drax, Endesa, Fortum, SSAB) and the new list includes small specialised emitters (REA Holdings, AEP Plantations, Pharos Energy) with effectively zero VW weight. This rotation explains why the VW benchmark itself decarbonised over the sample.

Rank	Company	ISIN	Country	CI (tCO ₂ /M\$rev)	VW wgt (%)
<i>End-2013</i>					
1	Drax Group	GB00B1VNSX38	GB	7,356.0	0.06
2	Buzzi	IT0001347308	IT	5,670.9	0.03
3	Holcim	CH0012214059	CH	4,279.1	0.26
4	Heidelberg Materials	DE0006047004	DE	3,083.0	0.15
5	Fortum	FI0009007132	FI	2,491.2	0.21
6	RWE	DE0007037129	DE	2,405.9	0.22
7	SSAB A	SE0000171100	SE	1,736.6	0.02
8	Endesa	ES0130670112	ES	1,260.7	0.36
9	Séché Environnement	FR0000039109	FR	1,252.1	0.00
10	Intl. Hotels Group	GB00BHJYC057	GB	1,213.3	0.10
<i>End-2018</i>					
1	Buzzi	IT0001347308	IT	6,045.7	0.03
2	Holcim	CH0012214059	CH	4,601.3	0.28
3	Heidelberg Materials	DE0006047004	DE	3,697.3	0.14
4	Fortum	FI0009007132	FI	3,592.8	0.22
5	Evrax (suspended)	GB00B71N6K86	GB	3,302.7	0.10
6	SBM Offshore	NL0000360618	NL	2,608.9	0.03
7	RWE	DE0007037129	DE	2,376.0	0.14
8	Pharos Energy	GB00B572ZV91	GB	2,179.3	0.00
9	Ferrexpo	GB00B1XH2C03	CH	1,918.8	0.02
10	Yara International	NO0010208051	NO	1,496.6	0.12
<i>End-2024</i>					
1	Titan	BE0974338700	BE	4,102.6	0.03
2	Buzzi	IT0001347308	IT	4,043.3	0.06
3	AEP Plantations	GB0000365774	GB	3,224.3	0.00
4	Heidelberg Materials	DE0006047004	DE	2,965.0	0.19
5	Evrax (suspended)	GB00B71N6K86	GB	2,859.7	0.01
6	Holcim	CH0012214059	CH	2,403.7	0.47
7	REA Holdings	GB0002349065	GB	2,325.3	0.00
8	Pharos Energy	GB00B572ZV91	GB	2,166.3	0.00
9	RWE	DE0007037129	DE	1,738.1	0.18
10	L’Air Liquide	FR0000120073	FR	1,192.1	0.79

Table 7: Top-10 carbon-intensive EUR firms by CI (tCO₂e per M\$ revenue), at three snapshot dates. VW weights are computed at the same dates. The rotation in the list illustrates the structural decarbonisation of the EUR universe over the sample.

The more important column for portfolio CF is the VW weight. In 2024, eight of the ten highest-CI firms have benchmark weights below 0.4% and four are effectively zero. The CI ranking and the CF-impact ranking are very different things. Titan and Buzzi have the two highest carbon

intensities in the EUR sample, but together they represent less than 0.10% of the benchmark, so their contribution to portfolio CF is negligible. By contrast, L’Air Liquide ranks only 10th by CI but has the largest VW weight in the table (0.79%) – it is likely the largest contributor to the benchmark’s CF among these ten firms, not because its emissions per dollar of revenue are highest but because the portfolio holds a larger position in it.

This illustrates the difference between WACI and CF. WACI is mainly driven by firms with extremely high carbon intensity, whereas CF is driven more by large firms with moderate but meaningful emissions per dollar of market cap. An effective decarbonisation strategy must therefore focus on the latter.

5.3 MV(0.5) – Carbon-Constrained Minimum Variance

A carbon constraint is now added to the MV optimisation. The investor still targets minimum variance but must keep the portfolio’s CF at or below 50% of the unconstrained MV portfolio’s CF each year:

$$\min_{\alpha_Y} \alpha_Y' \Sigma_Y \alpha_Y \quad \text{s.t.} \quad \text{CF}_Y^{(\alpha)} \leq 0.5 \cdot \text{CF}_Y^{(P^{(mv)})}, \quad \sum_i \alpha_{i,Y} = 1, \quad \alpha_{i,Y} \geq 0 \quad \forall i.$$

The target is calibrated annually against the unconstrained MV portfolio’s own CF, not the benchmark. Because MV CF is itself volatile, the annual targets vary considerably – from 12.7 in 2013 to 290.1 in 2014 (the year of the large MV CF spike). The targets are feasible every year: the EUR universe always contains enough low-emission stocks. Table 8 confirms that the constraint binds exactly in all 12 years, with realized CF equal to 50% of MV CF to machine precision.

Year	Target = 0.5 · CF _{MV}	Realized CF	Reduction (%)	Slack
2013	12.73	12.73	50.00	≈ 0
2014	290.07	290.07	50.00	≈ 0
2015	61.25	61.25	50.00	≈ 0
2016	272.31	272.31	50.00	≈ 0
2017	157.07	157.07	50.00	≈ 0
2018	90.33	90.33	50.00	≈ 0
2019	17.37	17.37	50.00	≈ 0
2020	27.02	27.02	50.00	≈ 0
2021	18.93	18.93	50.00	≈ 0
2022	22.38	22.38	50.00	≈ 0
2023	58.44	58.44	50.00	≈ 0
2024	33.68	33.68	50.00	≈ 0

Table 8: MV(0.5) verification — realized CF vs the 50% reduction target, by year. Slack denotes (target – realized) and is at machine-precision level in every year, confirming the constraint binds exactly.

Composition shifts. Table 9 reports the main composition changes from MV to MV(0.5) at three snapshot dates.

Year	Direction	Company	ISIN	Country	Δw (pp)	CI
2013	Removed ↓	United Utilities Group	GB00B39J2M42	GB	−5.14	178.8
	Removed ↓	Devro (delisted 14/04/23)	GB0002670437	GB	−2.15	387.0
	Removed ↓	National Grid	GB00BDR05C01	GB	−2.11	389.1
	Reduced ↓	Roche Holding	CH0012032048	CH	−1.66	16.2
	Reduced ↓	Teva Pharmaceutical	US8816242098	IL	−1.41	51.8
	Added ↑	Swisscom ‘R’	CH0008742519	CH	+3.20	2.0
	Added ↑	Dignity (delisted 25/05/23)	GB00BRB37M78	GB	+2.28	61.5
	Added ↑	Coloplast B	DK0060448595	DK	+2.01	21.0
	Added ↑	Chubb	CH0044328745	CH	+1.87	3.0
	Added ↑	Swedish Match (delisted)	SE0015812219	SE	+1.87	27.7
2018	Removed ↓	National Grid	GB00BDR05C01	GB	−2.22	334.0
	Removed ↓	Evraz (suspended 06/12/23)	GB00B71N6K86	GB	−1.63	3,302.7
	Reduced ↓	Perrigo	IE00BGH1M568	IE	−0.69	30.0
	Reduced ↓	Coloplast B	DK0060448595	DK	−0.64	20.6
	Reduced ↓	Compass Group	GB00BD6K4575	GB	−0.60	4.3
	Added ↑	bioMérieux	FR0013280286	FR	+1.06	23.2
	Added ↑	Swisscom ‘R’	CH0008742519	CH	+1.05	6.0
	Added ↑	Coca-Cola HBC	CH0198251305	CH	+0.98	77.7
	Added ↑	Johnson Controls Intl.	IE00BY7QL619	IE	+0.97	33.9
	Added ↑	United Utilities Group	GB00B39J2M42	GB	+0.81	142.6
2024	Removed ↓	Camellia	GB0001667087	GB	−2.54	582.8
	Reduced ↓	Swisscom ‘R’	CH0008742519	CH	−0.57	4.2
	Reduced ↓	Bezeq Israeli Telecom	IL0002300114	IL	−0.41	53.9
	Reduced ↓	Ebro Foods	ES0112501012	ES	−0.37	81.4
	Reduced ↓	Beiersdorf	DE0005200000	DE	−0.25	15.2
	Added ↑	Valiant ‘R’	CH0014786500	CH	+0.84	1.4
	Added ↑	Diageo	GB0002374006	GB	+0.61	26.3
	Added ↑	First Intl. Bk. of Israel	IL0005930388	IL	+0.48	3.4
	Added ↑	Jyske Bank	DK0010307958	DK	+0.47	0.4
	Added ↑	Rhoen-Klinikum	DE0007042301	DE	+0.47	34.9

Table 9: Main composition changes from MV to MV(0.5) at three snapshot dates. Δw is the weight change in percentage points. CI is in tCO₂e/M\$ revenue.

2013 – a structural tilt within the low-volatility segment. At the start of the sample the MV portfolio is dominated by UK regulated utilities. The constraint forces large reallocations: United Utilities (−5.14 pp), Devro (−2.15 pp) and National Grid (−2.11 pp) – three firms with CIs between 179 and 389 – are removed almost entirely. The optimiser does not reach for growth or cyclicals; it stays inside the defensive segment and substitutes cleaner alternatives: Swisscom (CI 2.0, telecom infrastructure), Coloplast (21.0, medical devices), Chubb (3.0, insurance). The tilt is large but *narrow* in sector terms – the carbon constraint reshuffles within low-volatility names; it does not change the portfolio’s risk character.

2018 – the same pattern, in milder form. The MV portfolio still tilts toward UK utilities (National Grid −2.22 pp) and now also a heavy emitter, Evraz (CI 3,303), is partially removed. Replacements continue the substitution logic of 2013: bioMérieux (diagnostics), Swisscom, Coca-Cola HBC, Johnson Controls. The magnitude of the largest tilt is roughly half that of 2013, suggesting the unconstrained MV portfolio already starts from a cleaner position.

2024 – almost no tilt at all. By the end of the sample, the largest removal is −2.54 pp (Camellia, a single high-CI agribusiness conglomerate) and *every other change is below 1 pp*. The optimiser is no longer reshaping the portfolio – it is making surgical adjustments. The reason is that the unconstrained MV portfolio has itself decarbonised over the decade: the UK utilities that dominated MV in 2013–2018 have either exited the low-volatility set (rising leverage, regulatory shocks) or seen their own carbon intensities fall as the UK power mix shifted. By 2024 the MV portfolio is already close to a low-carbon allocation, and the 50% constraint is met with small,

idiosyncratic substitutions toward Swiss/Nordic banks (Valiant, Jyske), a Swiss insurance pair, and defensive consumer/healthcare names (Diageo, Rhoen-Klinikum).

Implication for the financial cost of decarbonisation. The small size of the 2024 tilts is *not* evidence that the constraint has become non-binding – it is binding every year (Table 8). It is evidence that the constraint has become *cheap*, because the market itself has moved. This is consistent with the small Sharpe gap between MV and MV(0.5) observed in later years and stands in contrast to the larger frictional cost visible at the start of the sample.

Portfolio	Arith. (%)	Geom. (%)	Vol (%)	Sharpe	Min (%)	Max (%)	MDD (%)
$P^{(mv)}$	4.62	3.76	13.65	0.211	−12.02	14.03	−23.92
$P^{(mv)}(0.5)$	4.58	3.73	13.55	0.209	−11.39	13.42	−24.41
$P^{(vw)}$	8.19	7.16	15.87	0.406	−15.34	18.27	−28.17

Table 10: Performance comparison: MV baseline, CF-constrained MV(0.5), and VW benchmark, January 2014 to December 2025.

The financial impact of halving the CF is very small. Annualised arithmetic return decreases only from 4.62% to 4.58%; volatility decreases slightly from 13.65% to 13.55%; Sharpe drops from 0.211 to 0.209. These differences are economically negligible.

This is not a coincidence – it reflects a structural feature of the EUR universe. The MV optimiser mainly selects defensive stocks with low variance, such as utilities, telecoms, healthcare, and consumer staples. Within this defensive segment, higher-emission firms (utilities, some industrials) are often replaceable by similarly defensive but cleaner alternatives (telecoms, medtech, insurers). The carbon constraint and the financial objective therefore point in the same direction – away from volatile emitters and toward stable, lower-emission firms – which is why the constraint has limited impact: the unconstrained MV portfolio is already close to the carbon-constrained frontier.

The downside-risk profile confirms this interpretation. Max drawdown moves only from −23.92% (MV) to −24.41% (MV(0.5)), and the worst monthly return moves from −12.02% to −11.39%. The constrained portfolio does not become more fragile during market stress. This distinction matters: a decarbonised portfolio would be less attractive if the reduction in emissions came with hidden downside risk; here the carbon constraint reduces emissions without materially increasing financial vulnerability.

From a carbon perspective the constraint achieves exactly what it is designed to do. MV(0.5) tracks the annual target of 50% of the unconstrained MV footprint exactly, confirming the constraint is binding and respected across the sample. By 2024 MV(0.5) reaches CF of 33.68 tCO₂e/M\$ invested, compared with 67.36 for MV and 86.64 for VW. Although the optimisation constraint is imposed on CF rather than directly on WACI, the constrained portfolio also remains carbon-efficient in revenue-intensity terms (its WACI in 2024 is roughly 25 tCO₂e/M\$ revenue, vs 39.84 for MV and 84.55 for VW).

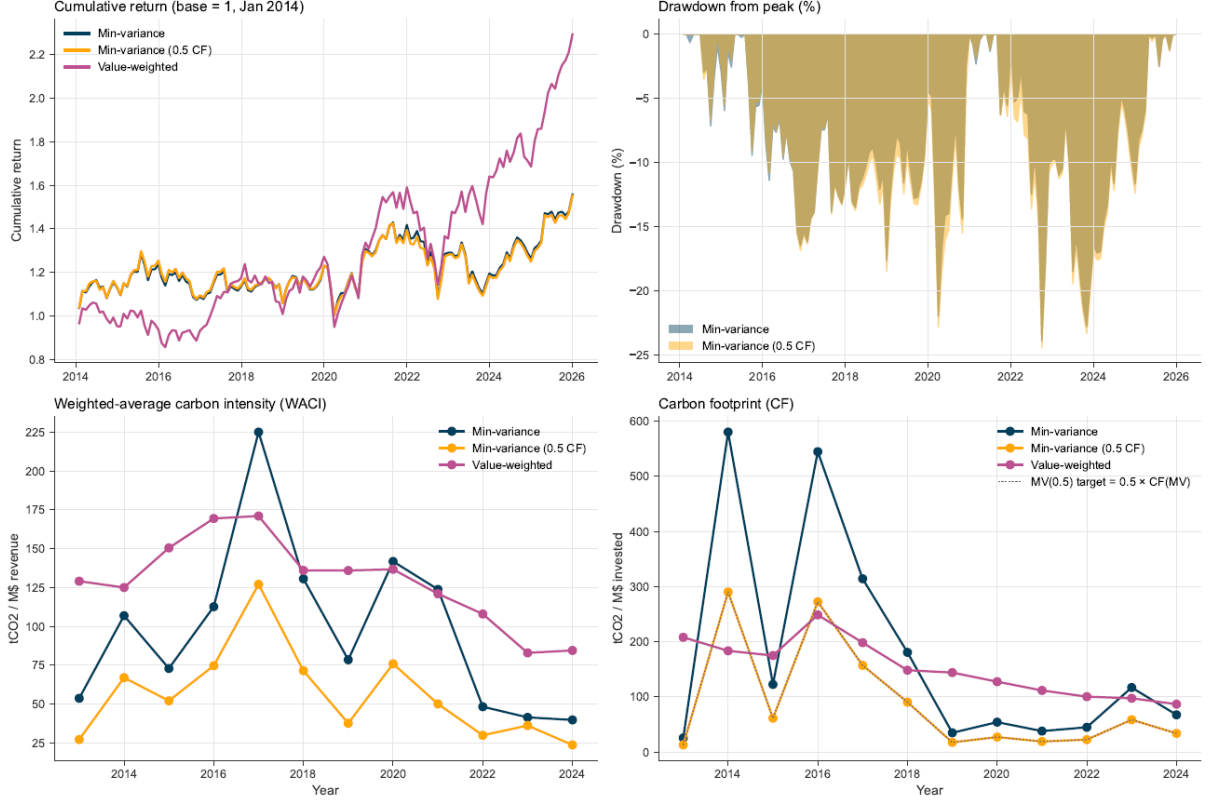


Figure 4: Min-Variance vs Min-Variance(0.5) vs VW, 2014–2025. Top-left: cumulative return. Top-right: CF with the 0.5-CF(MV) target overlaid. Bottom-left: WACI. Bottom-right: rolling realised volatility.

Overall, Figure 4 highlights the main result of Section 5.3: in this EUR sample, active decarbonisation through a MV framework is almost costless because the financial and carbon objectives are partially aligned. The optimiser can halve the portfolio’s CF without materially changing its cumulative performance, volatility, Sharpe ratio, or drawdown profile. For an active low-risk investor, carbon reduction can be implemented through targeted substitutions rather than a wholesale change in portfolio strategy.

5.4 VW(0.5) – Tracking-Error Minimisation with a CF Constraint

We now consider a passive investor who wants to reduce the portfolio’s CF while staying as close as possible to the VW benchmark. Unlike the MV strategy, the objective is not to minimise total portfolio risk but to minimise deviation from the benchmark.

At each rebalancing year Y , we solve:

$$\min_{\alpha_Y} (\alpha_Y - \alpha_Y^{(vw)})' \Sigma_Y (\alpha_Y - \alpha_Y^{(vw)}) \quad \text{s.t.} \quad \text{CF}_Y^{(\alpha)} \leq 0.5 \cdot \text{CF}_Y^{(vw)}, \quad \sum_i \alpha_{i,Y} = 1, \quad \alpha_{i,Y} \geq 0 \quad \forall i,$$

where $\alpha_Y^{(vw)}$ is the VW benchmark vector at the end of year Y and Σ_Y is the covariance matrix estimated using the rolling 10-year window. The portfolio remains benchmark-oriented but introduces an explicit decarbonisation objective.

Portfolio	Arith. (%)	Geom. (%)	Vol (%)	Sharpe	Min (%)	Max (%)	MDD (%)	TE (%)
$P^{(vw)}$	8.19	7.16	15.87	0.406	-15.34	18.27	-28.17	0.00
$P^{(vw)}(0.5)$	8.13	7.11	15.79	0.404	-15.22	18.21	-28.20	0.43
$P^{(mv)}(0.5)$	4.58	3.73	13.55	0.209	-11.39	13.42	-24.41	9.81

Table 11: Performance: VW benchmark, TE-minimising VW(0.5), and MV(0.5) reference, January 2014 to December 2025. Realized TE is the annualised standard deviation of monthly active returns ($R_p - R_{vw}$) times $\sqrt{12}$.

Implied Information Ratio for VW(0.5) is $(8.13 - 8.19) / 0.43 \approx -0.13$, economically indistinguishable from zero. The 50% CF constraint is binding every year (verified to machine precision in the notebook): realized CF equals $0.5 \cdot CF_{VW}$ in all 12 years.

Figure 5 shows the trade-off between carbon reduction and benchmark tracking. The VW(0.5) portfolio achieves a substantial decrease in CF while maintaining a financial profile close to the VW benchmark. Its cumulative return tracks VW for most of the sample, with a small performance gap that becomes visible after 2018 and remains until the end of the period.

Over the full period, annualised return decreases marginally from 8.19% for VW to 8.13% for VW(0.5). Volatility is essentially unchanged at 15.79% (vs 15.87%). Sharpe is stable at 0.404 vs 0.406. The 50% CF constraint is therefore achieved at minimal risk-adjusted cost in the EUR universe.

These results show that a 50% reduction in CF is feasible for a benchmark-oriented investor at a small but non-zero cost. As the VW benchmark holds large firms in proportion to market cap regardless of their emissions profile, the carbon constraint requires the optimiser to underweight some large emitters and reallocate capital toward cleaner firms. This adjustment meets the carbon objective but introduces a small, persistent deviation from the benchmark, which explains the slightly lower risk-adjusted performance.

The cost can be explained by the structure of the VW benchmark. Because firms are weighted by market cap, the benchmark naturally includes large companies independently of their carbon profile. Some firms therefore contribute substantially to portfolio CF simply because they represent a large share of the benchmark, even if they are not the most carbon-intensive in terms of revenue. To reduce CF by 50%, the optimiser underweights these large emitters and reallocates capital toward cleaner firms. This creates an active deviation from the benchmark which the tracking-error objective minimises but cannot fully eliminate. The small negative cumulative active return is the opportunity cost of reducing ownership-based emissions while staying benchmark-oriented.

By 2024 the VW(0.5) CF reaches 43.32 tCO₂e/M\$ invested (vs 86.64 for the unconstrained VW benchmark) and WACI falls to approximately 44 tCO₂e/M\$ revenue (vs 84.55 for VW). Although the constraint is on CF rather than WACI, the portfolio also becomes less carbon-intensive on a revenue basis – the optimisation shifts the allocation toward firms with cleaner operating profiles.

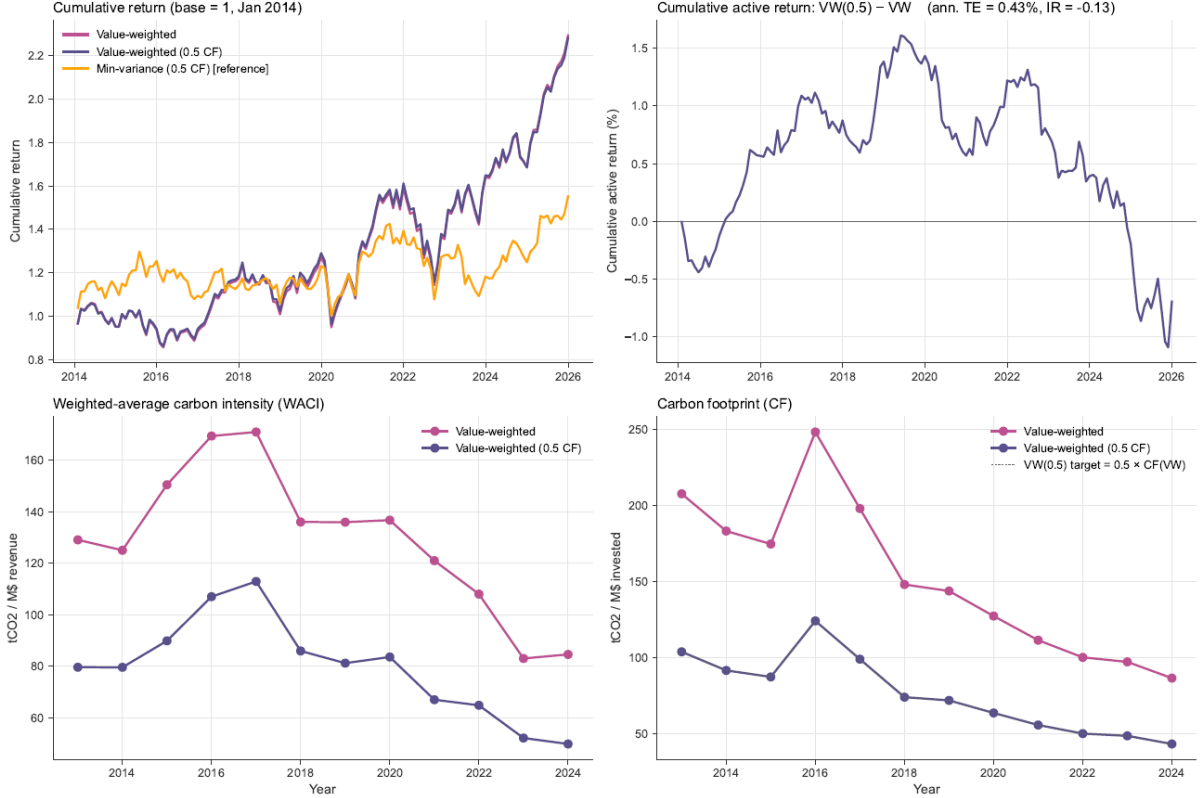


Figure 5: Financial and carbon performance of the VW(0.5) tracking-error strategy, 2014–2025. Top-left: cumulative return. Top-right: CF with the 0.5-CF(VW) target. Bottom-left: WACI. Bottom-right: cumulative active return vs VW with realized TE annotated.

5.5 VW(NZ) – Net-Zero Glide Path and Strategy Comparison

The comparison between the MV- and VW-based strategies highlights that the financial cost of decarbonisation depends strongly on the investor’s starting point. When the portfolio is built from a MV objective, the carbon constraint has almost no effect on performance. When the portfolio starts from the VW benchmark, the same 50% CF reduction is achieved at a small but measurable cost (approximately 6 bps annual return drag, $IR \approx -0.13$).

For the MV strategy, this result is mainly explained by the composition of the unconstrained MV portfolio: the optimiser already avoids the largest emitters in the EUR sample for financial reasons (cyclical industrials and energy companies tend to be more volatile, which makes them unattractive for variance minimisation). Adding a 50% CF constraint therefore only requires small, targeted substitutions within an already defensive portfolio, replacing a few utility or industrial positions with similarly low-volatility but cleaner telecom, medtech, or financial names. Sharpe barely changes (0.211 to 0.209) and the 12-year cumulative return paths are visually indistinguishable.

The VW strategy leads to a different conclusion. Because VW weights stocks in proportion to market cap, it is indifferent to carbon exposure and naturally holds large emitters when they represent large firms in the market. A 50% CF reduction forces the optimiser to underweight these firms relative to the benchmark, creating a systematic difference from VW with no perfect substitute to replace their role. The result is a return drag of approximately 6 basis points per year with realized TE of 0.43% annualised and $IR \approx -0.13$. The constraint is binding but does not materially affect risk-adjusted performance.

The asymmetry between the two cases – near-zero cost for MV, small but real cost for the benchmark-oriented strategy – is the central empirical finding of this section. It is explained

by the structure of the EUR investment universe: the overlap between firms that are financially unattractive to the MV optimiser and firms that are emissions-intensive is strong, so financial and carbon objectives naturally align for the MV strategy.

Net-zero glide path. Having compared the static 50% strategies, we now examine whether a more progressive decarbonisation path leads to a better balance between carbon reduction and financial performance. The net-zero strategy replaces the fixed annual 50% target with a glide path that reduces CF by 10% per year from the 2013 VW baseline:

$$CF_Y^{\text{target}} = (0.90)^{Y-2012} \times CF_{2013}^{(vw)}, \quad CF_{2013}^{(vw)} = 207.92 \text{ tCO}_2\text{e/M\$ invested}.$$

The same TE-minimisation framework as in Section 5.4 is used, but the carbon constraint is defined by this declining annual target. By 2024 the target reaches 58.7 tCO₂e/M\$ invested, corresponding to a 71.8% reduction from the 2013 baseline.

A key feature of the EUR sample is that the glide-path constraint is *not* binding in all years. In four of the twelve years (2014, 2015, 2018, 2021) the VW benchmark’s own carbon intensity, after the TE-min optimisation, already satisfies the glide target without forcing a tilt – the largest slack is 6.6 in 2018 (where realized CF = 103.86 vs target 110.50) and 3.0 in 2021. In the other eight years the constraint binds tightly. This pattern reflects the fact that the EUR benchmark itself is decarbonising over the sample period; in some years the market does the job for the investor.

Portfolio	Arith. (%)	Geom. (%)	Vol (%)	Sharpe	Min (%)	Max (%)	MDD (%)	TE (%)
$P^{(vw)}$	8.19	7.16	15.87	0.406	−15.34	18.27	−28.17	0.00
$P^{(vw)}(0.5)$	8.13	7.11	15.79	0.404	−15.22	18.21	−28.20	0.43
$P^{(vw)}(\text{NZ})$	8.18	7.17	15.80	0.407	−15.12	18.23	−28.40	0.38

Table 12: Performance: VW benchmark, VW(0.5) (50% flat target), and VW(NZ) (glide path). January 2014 – December 2025. Realized TE is annualised standard deviation of monthly active returns.

Implied IRs are approximately −0.13 for VW(0.5) and −0.01 for VW(NZ), both economically indistinguishable from zero. Notably, the VW(NZ) Sharpe (0.407) is essentially identical to the benchmark’s (0.406) – the net-zero strategy is not financially worse than the unconstrained benchmark over this sample, despite delivering a 71.8% reduction in CF by 2024.

While Table 12 summarises average performance, Figure 6 shows how the financial and carbon profiles evolve over time. The cumulative return panel shows that both VW(0.5) and VW(NZ) remain close to VW; a performance gap is visible toward the end of the period but remains moderate.

The CF panel is the central result of the net-zero analysis. The VW benchmark already declines over time, but the two constrained strategies impose stronger carbon discipline. VW(0.5) stays close to half of VW CF each year, whereas VW(NZ) follows the $(1-\theta)^k$ glide path. The net-zero path is less restrictive than VW(0.5) at the start of the sample but becomes stricter over time and reaches around 58.7 tCO₂e/M\$ invested by 2024 (vs 43.3 for VW(0.5) and 86.6 for VW). The WACI and active-return panels confirm the same picture: both constrained portfolios are less carbon-intensive than VW and track it very closely.

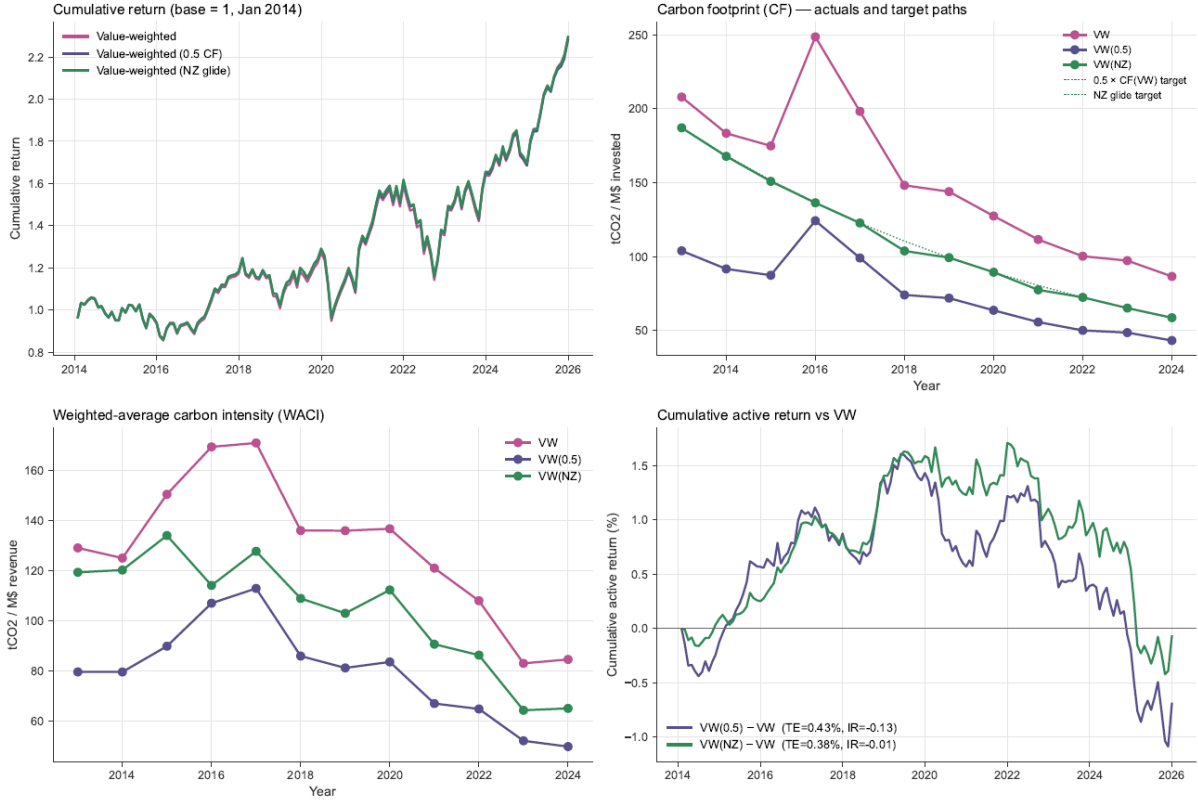


Figure 6: Net-zero glide path strategy compared with VW and VW(0.5), 2014–2025. Top-left: cumulative return. Top-right: CF with both target paths. Bottom-left: WACI. Bottom-right: cumulative active return vs VW with realized TE and IR annotated.

Overall, the net-zero strategy achieves a credible and predictable reduction in CF while remaining financially close to VW. Compared with VW(0.5), VW(NZ) is less aggressive at the beginning but becomes stricter over time, making it more aligned with a long-term decarbonisation objective. The financial cost is essentially nil at the full-period horizon (Sharpe 0.407 vs 0.406), although the sub-period evidence shows a moderate active drag in the middle of the sample.

6 Suggestions for Improvements and Robustness Checks

Although the implementation follows the project requirements, several extensions could improve robustness and practical relevance.

Covariance estimator. The current code estimates a pairwise-complete sample covariance matrix and projects it onto the PSD cone. This is appropriate because the raw matrix can be indefinite due to missing-data handling and because the number of firms (around 600) is much larger than the 120 monthly observations in the rolling window. A useful robustness check would compare this approach with (i) a covariance matrix computed only on months with complete data for all firms, (ii) Ledoit-Wolf shrinkage, and (iii) a factor-model covariance. Because the MV portfolio is highly sensitive to Σ in the low-data regime, these alternatives would primarily affect the MV/MV(0.5) results in Tables 6 and 10; the VW-family results are less exposed because the benchmark is fixed.

Concentration constraints. The MV portfolio is highly concentrated, with 20–32 active positions out of several hundred eligible firms. A robustness check would impose a maximum weight per firm (e.g., 5%) or a minimum number of positions. This would make the portfolio more realistic for institutional asset management and reduce idiosyncratic risk. The reported MV Sharpe of 0.211 is likely to fall further under such constraints (since the optimiser is forced

away from its preferred low-variance subset), but the carbon-cost gap MV vs MV(0.5) is likely to remain small for the same structural reason discussed in Section 5.3.

Sector neutrality. The MV allocation tilts toward defensive, low-volatility sectors and may therefore become strongly exposed to specific industries. Adding sector-neutrality constraints relative to the VW benchmark would help to distinguish whether performance differences are due to the optimisation itself or to sector tilts. For the VW(0.5) and VW(NZ) strategies, sector neutrality would directly affect the realized TE – making it harder to maintain the carbon constraint but more reassuring from a risk-decomposition perspective.

Transaction costs and turnover. The current implementation rebalances active portfolios annually and allows weights to drift within the year. Turnover is not quantified or penalised. Adding turnover statistics (e.g., $\tau_Y = \frac{1}{2} \sum_i |\alpha_{i,Y} - \alpha_{i,Y-1}^{\text{drift}}|$) and modelling proportional trading costs would make the backtest more realistic and would show whether the carbon-constrained and net-zero portfolios require substantially more rebalancing than the unconstrained portfolios. From Table 9, MV(0.5) appears to involve large reallocations early in the sample (5+ pp tilts in 2013) and surgical adjustments later (sub-1 pp in 2024); a turnover series would quantify this transition.

Scope-3 sensitivity. Scope 3 emissions are not included in this analysis (per the project brief) but can dominate total climate exposure in sectors such as financials, retail, technology, and consumer goods. A robustness check using Scope 1+2+3 (where available) would likely tighten the carbon constraint substantially for VW-based strategies (since many financial and tech names that look clean on Scope 1+2 carry large Scope 3 emissions through their portfolios or supply chains).

7 Limitations

- **Scope 1+2 only.** The carbon analysis relies only on Scope 1 and Scope 2 emissions. Although these capture direct and energy-related emissions, they exclude Scope 3, which can represent the bulk of climate exposure in sectors such as financials, retail, technology, and consumer goods. The conclusion that the EUR universe contains plentiful low-carbon substitutes (Section 5.3) may not hold under Scope 3 accounting. In addition, carbon and revenue data may not be perfectly comparable across firms because reporting standards, boundaries, and disclosure quality differ. When data are missing we carry forward the last available value, which improves coverage but creates stale observations that may not reflect the firm’s most recent environmental profile.
- **Parameter sensitivity.** Several choices affect the universe and the allocations: the minimum number of valid return observations (36), the stale-price threshold (30%), the 10-year estimation window, and the treatment of very low return-index values ($\text{LOW_FLOOR} = 0.5$). A stricter stale-price filter improves liquidity screening but may exclude eligible firms; a longer estimation window would smooth the covariance but slow adaptation to regime shifts. The MV concentration in Table 5 and the MV Sharpe in Table 6 are the results most sensitive to these choices.
- **Covariance estimation noise.** The covariance matrix is estimated using 120 monthly observations for ~600 firms. Covariance estimates are therefore noisy even after PSD projection. Small changes in Σ can lead to meaningful changes in MV weights, which is the standard caveat of the MV strategy. The TE-minimising portfolios are less exposed to this issue because they anchor on $\alpha^{(vw)}$.
- **Portfolio concentration.** The MV portfolio holds only 20–32 names. While concentration may reduce estimated volatility, it can also increase exposure to firm-specific risk and is unre-

alistic for institutional asset management. Adding a maximum weight per firm or a minimum number of holdings would address this but is beyond the scope of the project brief.

- **Generalisability.** The result that carbon reduction is essentially free for MV-based strategies and cheap for VW-based strategies is specific to the EUR region and to 2014–2025. In another market or time period, high-emission firms could also be important low-volatility or benchmark-heavy constituents, making decarbonisation more costly. The qualitative ranking of strategies is likely to be robust but the quantitative gaps (e.g., 6 bps drag for VW(0.5)) should not be extrapolated.

Use of Large Language Models (LLMs)

In accordance with the project guidelines, this section documents our use of generative AI tools. We used Claude (Anthropic) and ChatGPT (OpenAI) as supporting tools during the project. Their use was limited to technical assistance, including: (i) debugging Python code, especially indexing issues, portfolio-weight alignment, and optimisation errors; (ii) understanding error messages from `scipy.optimize.minimize`, including convergence warnings and constraint feasibility issues; (iii) improving code readability through clearer comments, docstrings, and function organisation; and (iv) polishing language and structure of this report (clarity and grammar). All numerical results were produced by running our own Python notebook. No AI tool was used to generate the project solution end-to-end. The interpretation of the results, the discussion of limitations, and the final conclusions reflect the group's own analysis. The group remains fully responsible for the correctness, reproducibility, and academic integrity of this report.